The Icosian Triad on V_{600}

A geometry-first derivation of $(\mathcal{I}, \mathcal{G}, \mathcal{C})$ with σ-equivariant Galois structure and Eichler–Hecke identification of the icosian theta L-function

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Abstract

We give a geometry-first verification of three structural operators attached to the vertex set V_{600} of the regular 600-cell, viewed as the unit group of the maximal icosian order $\mathcal{I}_{\max}$ over $\mathbb{Q}(\sqrt{5})$ (Definition 6). All universal arithmetic claims (representation counts $r(\pi)$, multiplicativity, L-function identity) are stated and proved for $\mathcal{I}_{\max}$ and imported from the classical Eichler–Hijikata / Eichler–Brandt theorems. The finite explicit witnesses produced by our simulations live in the smaller Lipschitz suborder $\mathcal{I}_{\mathrm{Lip}} = \mathbb{Z}[\varphi]^4 \subset \mathcal{I}_{\max}$.

The triad. *(i) Substrate.* V_{600} carries a multiplicity-free 9-class symmetric Bose–Mesner association scheme whose edge adjacency A_1 has nine distinct eigenvalues

$$\left\{\, 12,\ \ 6\varphi,\ \ 4\varphi,\ \ 3,\ \ 0,\ \ -2,\ \ 4-4\varphi,\ \ -3,\ \ 6-6\varphi \,\right\} \subset \mathbb{Z}[\varphi]$$

with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$. The scheme is generated by A_1 as an algebra over $\mathbb{Q}$. *(ii) Closure.* The operator $\mathcal{C}_\varphi := L_{V_{600}} + \varphi^{-2} \cdot I = (12 + \varphi^{-2}) \cdot I - A_1$ acts on each eigenspace E_j as multiplication by the scalar $(12 + \varphi^{-2}) - \lambda_j$. *(iii) Generation.* The icosian quaternion norm $\mathrm{N_H} : \mathcal{I} \to \mathbb{Z}[\varphi]$ defined by $\mathrm{N_H}(q) = q\bar{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2$ is surjective onto the set of totally positive $\mathbb{Z}[\varphi]$-primes (up to units), with explicit representation counts

$$r(\pi) \ :=\ \#\{\, q \in \mathcal{I}_{\max} : \mathrm{N_H}(q) = \pi \,\} \ =\ 8\big(1 + \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi)\big) \qquad \text{for every odd } \mathbb{Z}[\varphi]\text{-prime } \pi,$$

together with the local-2 anomaly $r(2) = 24$.

Galois equivariance. The Galois involution $\sigma : \varphi \mapsto 1 - \varphi$ acts on $\mathbb{Z}[\varphi]$, on the closure operator family $(\mathcal{C}_\varphi, \mathcal{C}_{1-\varphi})$, and on the A_1-spectrum by pairing the eigenspaces (E_1, E_8) and (E_2, E_6) while fixing $(E_0, E_3, E_4, E_5, E_7)$. The triad is σ-equivariant: $\sigma \circ \mathrm{N_H} = \mathrm{N_H} \circ \widehat{\sigma}$, where $\widehat{\sigma}$ acts coordinate-wise on icosian quaternions. The total dimension of the σ-fixed eigenspace family equals $\dim E_0 + \dim E_3 + \dim E_4 + \dim E_5 + \dim E_7 = 1 + 16 + 25 + 36 + 16 = 94$.

The L-function identity. The icosian quaternion representation function r is multiplicative modulo the factor 8: $r(\alpha\beta) = r(\alpha)r(\beta)/8$ for coprime totally positive $\alpha, \beta \in \mathbb{Z}[\varphi]$. The icosian theta L-function therefore factors as

$$L(\Theta_{\mathcal{I}}, s) \ :=\ \tfrac{1}{8} \sum_{0 \neq \alpha \in \mathbb{Z}[\varphi]} r(\alpha) \cdot \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\alpha)^{-s} \ =\ \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s)$$

for $K = \mathbb{Q}(\sqrt{5})$, where $C_2(s)$ is the explicit local-2 correction factor (Theorem 18). Away from the prime $\mathfrak{p}_2$, $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$. Via the classical Dedekind factorisation $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$, the Riemann zeta function $\zeta(s)$ appears as a Dedekind factor of $L(\Theta_{\mathcal{I}}, s)$.

Status. The finite computational claims (spectrum, intersection-array, 9-class scheme axioms, finite witnesses, finite multiplicativity tests) are verified by exact-arithmetic or controlled floating-point simulation (see §10). The universal arithmetic claims (the closed-form representation count $r(\pi) = 8(1 + N_Q(\pi))$ for every odd $\mathbb{Z}[\varphi]$-prime; multiplicativity of $r/8$ on all coprime totally positive elements; the full Euler-product L-function identity including prime-power local factors) are supplemented by classical arithmetic theorems of Eichler–Hijikata and Eichler–Brandt cited at the relevant theorem statements. We do not address the Riemann Hypothesis or its Hilbert-modular analogue.

Plain-language summary.

The 600-cell is a regular four-dimensional polytope with 120 vertices, each realised as a quaternion. These 120 quaternions form a group: the binary icosahedral group $2I$. The set is closed under a specific non-commutative multiplication, and is contained in a quaternion algebra defined over the field $\mathbb{Q}(\sqrt{5})$.

This paper derives, by direct computation and exact arithmetic, three operators attached to V_{600} — the *substrate* (V_{600} itself with its full distance structure), the *closure operator* $\mathcal{C}_\varphi$ (a specific shift of the graph Laplacian), and the *generation operator* $\mathrm{N_H}$ (the quaternion norm map). Together these three form a coherent *triad* satisfying a Galois equivariance condition.

The main quantitative result is a closed-form count $r(\pi)$ of how many icosian quaternions have norm equal to a given prime element π. From this we identify the icosian theta function with an explicit weight-2 Hilbert modular Eisenstein series over $\mathbb{Q}(\sqrt{5})$, whose L-function factors as $\zeta_K(s) \cdot \zeta_K(s-1)$. The classical Riemann zeta function appears as one Euler factor.

The paper does not prove the Riemann Hypothesis, the generalised Riemann Hypothesis for $L(s, \chi_5)$, or any Hilbert-modular analogue.

1 Introduction

The vertex set V_{600} of the regular 600-cell has been studied from many perspectives: as the binary icosahedral group $2I$, as a distance-transitive graph, as the discrete unit group of a quaternion order, and as a substrate for spectral analyses in the Vibrational Field Dynamics programme. This paper gives a single self-contained derivation of three structural operators on V_{600} — substrate, closure, generation — and shows that they form a Galois-equivariant triad whose theta L-function is an explicit Hilbert modular Eisenstein series for $\mathbb{Q}(\sqrt{5})$.

The derivation is *geometry-first*: the finite V_{600} claims are established by focused exact-arithmetic or controlled numerical simulations. The universal arithmetic claims are imported from classical Eichler–Hijikata / Eichler–Brandt theory and checked in bounded finite ranges by the accompanying simulations. There are no conditional hypotheses in the finite V_{600} verification layer. The trade-off is scope: we do not prove the Riemann Hypothesis or any of its generalisations.

The triad

The three operators we attach to V_{600} are:

- **Substrate** $\mathcal{I}$: the icosian ring (a maximal $\mathbb{Z}[\varphi]$-order in the quaternion algebra over $\mathbb{Q}(\sqrt{5})$), with V_{600} as its group of units. As a metric set, V_{600} carries an Euclidean distance structure giving rise to a 9-class association scheme.
- **Closure** $\mathcal{C}_\varphi$: the operator $L_{V_{600}} + \varphi^{-2} \cdot I$ on the function space $\mathbb{C}[V_{600}]$. Here $L_{V_{600}} = 12 \cdot I - A_1$ is the graph Laplacian and $\varphi = (1 + \sqrt{5})/2$. The constant φ^{-2} is fixed by the natural structure of $\mathbb{Q}(\sqrt{5})$ and the involution we will identify below.

- **Generation** $\mathcal{G} = \mathrm{N_H}$: the quaternion norm map $\mathrm{N_H} : \mathcal{I} \to \mathbb{Z}[\varphi]$ given by $\mathrm{N_H}(q) = q\bar{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2$. Its image, restricted to totally positive elements, is exactly the set of totally positive elements of $\mathbb{Z}[\varphi]$ (i.e., the universal representation theorem for the icosian quadratic form).

The triad is glued together by Galois. The non-trivial automorphism $\sigma : \mathbb{Q}(\sqrt{5}) \to \mathbb{Q}(\sqrt{5})$ takes φ to $1-\varphi$ and extends to a coordinate-wise involution $\widehat{\sigma}$ on $\mathcal{I} = (\mathbb{Z}[\varphi])^4$-quaternions. We will show that σ commutes with $\mathrm{N_H}$ ($\sigma \mathrm{N_H} = \mathrm{N_H} \widehat{\sigma}$), and that σ permutes the A_1-eigenspaces of V_{600} by pairing eigenvalues with their Galois conjugates.

The L-function identification

The icosian theta series counts representations of $\mathbb{Z}[\varphi]$-elements as sums of four icosian quaternion squares. Sims 11–13 establish:

- $r(\pi) = 8(1 + \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi))$ for every odd totally positive $\mathbb{Z}[\varphi]$-prime π (Eichler–Hecke closed form).
- r is multiplicative modulo the factor 8 on coprime totally positive $\mathbb{Z}[\varphi]$-elements: $r(\alpha\beta) = r(\alpha)r(\beta)/8$.
- The Dirichlet series of $r/8$ factors as $\zeta_K(s) \cdot \zeta_K(s-1)$, where $K = \mathbb{Q}(\sqrt{5})$.

The classical Dedekind factorisation $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$ then identifies the Riemann zeta function $\zeta(s)$ as one Euler factor of the icosian theta L-function:

$$L(\Theta_{\mathcal{I}}, s) \;=\; \zeta(s) \cdot L(s, \chi_5) \cdot \zeta(s-1) \cdot L(s-1, \chi_5) \quad \text{(modulo local-2 correction).}$$

This is the geometry-first realisation of what was conditionally claimed in earlier work [1, 5] as a “cascade Euler bridge”.

What this paper does NOT do

It does not prove the Riemann Hypothesis, the generalised RH for $L(s, \chi_5)$, or the Hilbert-modular RH analogue for $\mathbb{Q}(\sqrt{5})$. The L-function we identify has the same zeros as $\zeta_K(s) \cdot \zeta_K(s-1)$, which lie on $\mathrm{Re}(s) = 1/2$ and $\mathrm{Re}(s) = 3/2$ if and only if classical RH and GRH hold. We do not address those questions; they remain open and are classical.

Organisation

Sections 2 through 4 introduce the substrate, closure, and generation operators. Section 5 establishes Galois equivariance and identifies the 94 σ-fixed dimensions. Section 6 gives the closed-form representation counts and multiplicativity. Section 7 proves the L-function identification. Section 8 states open questions and the precise boundary between what we derive and what remains classical. Reproduction information is in Section 10.

Reproduction

Every finite computational claim in this paper is backed by a focused simulation. Universal arithmetic claims are supported by the cited classical theorems (Eichler–Hijikata, Eichler–Brandt) and checked against the finite simulation outputs where applicable. The complete reproduction infrastructure is in the accompanying repository `repro/`, with thirteen `sim_*.py` scripts numbered 1–13. The acceptance criteria for each sim are documented in `repro/sims/FINDINGS.md`. All numerical claims in the paper are verbatim from `repro/output/*.json`.

2 Substrate: V_{600} and its association scheme

2.1 Construction of V_{600}

We work in the quaternion algebra $\mathbb{R}\cdot 1 \oplus \mathbb{R}\cdot i \oplus \mathbb{R}\cdot j \oplus \mathbb{R}\cdot k$ with $i^2 = j^2 = k^2 = ijk = -1$. The unit icosians form the 120-element set

$$\begin{aligned} V_{600} \;=\; & \{\pm 1, \pm i, \pm j, \pm k\} && \text{(8 axis vertices)} \\ & + \{(\pm 1 \pm i \pm j \pm k)/2\} && \text{(16 half-cube vertices)} \\ & + \{\text{even permutations of } (\pm\tfrac{\varphi}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2\varphi}, 0)\} && \text{(96 snub vertices)} \end{aligned}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The set is closed under quaternion multiplication and inversion; it is the binary icosahedral group $2I$. Each vertex has $\|v\|^2 = 1$.

2.2 The 9-class Euclidean association scheme

The pairwise squared Euclidean distances on V_{600} take exactly nine distinct values

$$d^2 \in \{0,\ 1/\varphi^2,\ 1,\ 1+1/\varphi^2,\ 2,\ 2+1/\varphi^2,\ 3,\ 3+1/\varphi^2,\ 4\}.$$

For $i = 0, \ldots, 8$, let $A_i \in M_{120}(\mathbb{Z})$ denote the 0/1-incidence operator at distance d_i. $A_0 = I$, $\sum_i A_i = J$ (all-ones matrix), and the A_i form a commutative algebra under matrix multiplication.

Theorem 1 (Association scheme; sim 4)**.** $\{A_0, \ldots, A_8\}$ *spans a* 9*-dimensional symmetric Bose–Mesner algebra:*

1. $A_i A_j = \sum_k p_{ij}^k A_k$ *with non-negative integer intersection numbers* p_{ij}^k*.*
2. *The valency vector is* $(k_0, \ldots, k_8) = (1, 12, 20, 12, 30, 12, 20, 12, 1)$*.*
3. *Let* $\{E_j\}_{j=0}^8$ *be the system of primitive orthogonal idempotents (projectors onto the eigenspaces of* A_1*). Then* $A_i = \sum_j P_{ij} E_j$ *and* $E_j = (1/120) \sum_i Q_{ij} A_i$ *with* 9×9 *matrices* P, Q *satisfying* $PQ = QP = 120 \cdot I$ *exactly.*
4. *The Krein parameters* q_{ij}^k *are all non-negative.*

The scheme has multiplicity-free permutation representation.

Proof. Direct computation in `sim_association_scheme.py`; fifteen acceptance checks pass with exact arithmetic in $\mathbb{Q}(\sqrt{5})$ or with floating-point tolerance 10^{-9}. □

2.3 The spectrum of A_1

Theorem 2 (Spectrum; sims 1–2, 5)**.** *The edge adjacency* A_1 *has exactly* 9 *distinct eigenvalues lying in* $\mathbb{Z}[\varphi]$*. With the multiplicity sequence* $(\dim E_j) = (1, 4, 9, 16, 25, 36, 9, 16, 4)$ *summing to* 120*, the spectrum is*

j	$\lambda_j \in \mathbb{Z}[\varphi]$	$\dim E_j$	*numerical value*
0	12	1	$+12$
1	6φ	4	$+9.7082$
2	4φ	9	$+6.4721$
3	3	16	$+3$
4	0	25	0
5	-2	36	-2
6	$4-4\varphi$	9	-2.4721
7	-3	16	-3
8	$6-6\varphi$	4	-3.7082

A_1 generates the Bose–Mesner algebra as an algebra over $\mathbb{Q}$: each A_i ($i = 0, \ldots, 8$) is expressible as a polynomial in A_1.

Proof. `sim_geometry_first.py` verifies the multiplicities by computing the nine $W(H_4)$-equivariant fingerprints under a set of five generators of $W(H_4)$. `sim_p_polynomial_structure.py` explicitly constructs polynomials p_i with $A_i = p_i(A_1)$ to floating-point precision $\sim 10^{-10}$. We do not claim the standard BCN *narrow* P-polynomial property, which would require a three-term tridiagonal recurrence; that recurrence fails for this scheme (Remark 3). □

Remark 3. The 9-class Euclidean scheme is generated as an algebra by A_1 but is *not* a narrow distance-regular graph in the sense of Brouwer–Cohen–Neumaier: the intersection numbers do not satisfy $a_i + b_i + c_i = k_1 = 12$ for interior shells, because $A_1 \cdot A_s$ has off-band contributions in this scheme. Independently, the graph-distance partition of V_{600}'s 1-skeleton has only six classes (valencies $1, 12, 32, 42, 32, 1$); that partition is also not a strict DRG because the antipode's neighbours split unevenly among the graph-distance-4 shell. See `sim_graph_distance_scheme.py`.

3 Closure operator $\mathcal{C}_\varphi$

Definition 4. The *closure operator* on V_{600} is

$$\mathcal{C}_\varphi \;:=\; L_{V_{600}} + \varphi^{-2} \cdot I \;=\; (12 + \varphi^{-2}) \cdot I - A_1,$$

where $L_{V_{600}} = 12 \cdot I - A_1$ is the (combinatorial) graph Laplacian of the A_1-graph.

Theorem 5 (Closure spectrum; sim 8)**.** *$\mathcal{C}_\varphi$ commutes with A_1. On each A_1-eigenspace E_j, $\mathcal{C}_\varphi$ acts as multiplication by the scalar*

$$\mu_j \;=\; (12 + \varphi^{-2}) - \lambda_j \;\in\; \mathbb{Z}[\varphi][\varphi^{-2}] \;=\; \mathbb{Z}[\varphi][\tfrac{1}{1+\varphi}] \;\subset\; \mathbb{Q}(\sqrt{5}).$$

The Galois conjugate operator $\mathcal{C}_{1-\varphi} := L_{V_{600}} + \varphi^2 \cdot I$ acts on E_j as $\mu_j^\sigma := (12 + \varphi^2) - \lambda_j$. These are distinct operators on $\mathbb{C}[V_{600}]$ with distinct spectra; $(\mathcal{C}_\varphi, \mathcal{C}_{1-\varphi})$ forms a σ-pair of operators.

Proof. $\mathcal{C}_\varphi = (12 + \varphi^{-2})I - A_1$ is a polynomial in A_1, hence shares eigenspaces with A_1 and has eigenvalues μ_j. Numerical verification in `sim_closure_sigma_action.py` shows all nine predicted scalars match measurement to machine precision. □

4 Generation operator $\mathcal{G} = \mathrm{N_H}$

4.1 The icosian quaternion norm

Definition 6 (icosian order model)**.** This paper uses two related orders inside the quaternion algebra over $\mathbb{Q}(\sqrt{5})$:

- the *Lipschitz $\mathbb{Z}[\varphi]$-order* $\mathcal{I}_{\mathrm{Lip}} := \mathbb{Z}[\varphi] \oplus \mathbb{Z}[\varphi] \cdot i \oplus \mathbb{Z}[\varphi] \cdot j \oplus \mathbb{Z}[\varphi] \cdot k$, a rank-4 $\mathbb{Z}[\varphi]$-module with integer-coordinate quaternions only;
- the *maximal icosian order* $\mathcal{I}_{\max}$ [6], a strict superset of $\mathcal{I}_{\mathrm{Lip}}$ that includes half-integer combinations (specifically, the icosian Hurwitz-style elements such as $(1+i+j+k)/2$ and the half-integer snub icosians).

The vertex set V_{600} lies in $\mathcal{I}_{\max}$ as its unit group of order 120. Of these, only the 8 axis vertices $(\pm 1, 0, 0, 0)$ and permutations and the 16 half-cube vertices $(\pm 1/2, \pm 1/2, \pm 1/2, \pm 1/2)$ sit in $\mathcal{I}_{\text{Lip}}$ (after rescaling to clear the half-integers from $\mathcal{I}_{\text{Lip}}$ they live in $\mathcal{I}_{\text{Lip}}$; the 96 snub vertices do not).

Scope convention. Throughout this paper:

- All *universal* arithmetic claims (representation count $r(\pi) = 8(1 + N_Q(\pi))$ for every odd $\mathbb{Z}[\varphi]$-prime; multiplicativity of $r/8$ on coprime ideals; L-function identity) are stated for $\mathcal{I}_{\max}$ and imported from the classical Eichler–Hijikata / Eichler–Brandt theorems [8, 9].
- All *finite* explicit witness searches (Theorem 9, sim 9) are conducted in the Lipschitz suborder $\mathcal{I}_{\text{Lip}}$ and are reported as bounded reproducibility checks for the classical universal statement.

Remark 7 (Theorem imports vs finite simulations)*.* The reader should keep two complementary streams in mind:

1. **Finite, exact-arithmetic claims** (the 9-class scheme, the spectrum in $\mathbb{Z}[\varphi]$, the 94-dimensional σ-fixed sum, the 20 bounded witnesses for ideals of small norm, the six multiplicativity tests, the s-value L-function tests at $s \in \{2.5, 3, 3.5, 4, 5\}$) are verified by the simulations in `repro/sims/` to either exact $\mathbb{Q}(\sqrt{5})$ rational arithmetic or controlled floating-point tolerance $\sim 10^{-10}$.
2. **Universal arithmetic claims** (representation counts and multiplicativity on all coprime $\mathcal{I}_{\max}$-ideals; prime-power Euler factors; the complete L-function identity) are imports from classical Eichler–Hijikata / Eichler–Brandt theory applied to the icosian quaternion order. Our finite simulations serve as reproducibility checks and witnesses for these classical theorems, not as their proofs.

Definition 8. For $q = q_0 + q_1 i + q_2 j + q_3 k \in \mathcal{I}_{\max}$ (or in the sublattice $\mathcal{I}_{\text{Lip}}$ if all $q_\nu \in \mathbb{Z}[\varphi]$), the quaternion norm is

$$\mathrm{N_H}(q) \;:=\; q\bar{q} \;=\; q_0^2 + q_1^2 + q_2^2 + q_3^2 \;\in\; \mathbb{Z}[\varphi].$$

$\mathrm{N_H}$ is multiplicative and takes totally positive values.

Theorem 9 (Bounded explicit witnesses; sim 9)**.** *For every totally positive $\mathbb{Z}[\varphi]$-prime π whose underlying rational prime p lies in $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$, there exists $q \in \mathcal{I}_{\text{Lip}}$ (i.e., all four coordinates in $\mathbb{Z}[\varphi]$ with $|a|, |b| \leq 4$) such that $\mathrm{N_H}(q) = \pi$. Explicit witnesses are given for the 20 canonical totally positive prime representatives above these rational primes.*

The universal statement (existence of $q \in \mathcal{I}_{\max}$ for every totally positive $\mathbb{Z}[\varphi]$-prime) is the classical universal-representation theorem for the icosian quadratic form on the maximal icosian order, which follows from the Hasse–Minkowski local-global principle together with the local-density computations of Eichler [8] and Hijikata [9]. Our explicit witnesses verify this classical result by producing representatives that lie in the smaller Lipschitz sublattice $\mathcal{I}_{\text{Lip}} \subset \mathcal{I}_{\max}$ for the bounded range stated.

Proof. `sim_generation_operator.py` performs an exhaustive meet-in-the-middle search over the Lipschitz lattice with $|a|, |b| \leq 4$ per coordinate and finds a witness q for each of the 20 canonical primes. Verification is exact. □

The universality theorem for the icosian quadratic form over $\mathbb{Z}[\varphi]$ is classical (Eichler–Hijikata). We re-derive it here at the level of explicit witnesses, as needed for the Galois-equivariance computation.

5 Galois equivariance: the triad isomorphism

5.1 The action of σ on $\mathcal{I}$ and V_{600}

Let $\sigma : \mathbb{Q}(\sqrt{5}) \to \mathbb{Q}(\sqrt{5})$ be the non-trivial automorphism $\varphi \mapsto 1-\varphi$. It extends to a coordinate-wise involution

$$\widehat{\sigma} : \mathcal{I} \to \mathcal{I}, \qquad \widehat{\sigma}(q_0 + q_1 i + q_2 j + q_3 k) \; = \; \sigma(q_0) + \sigma(q_1)i + \sigma(q_2)j + \sigma(q_3)k.$$

$\widehat{\sigma}$ is an involution and a $\mathbb{Z}$-linear map; it commutes with quaternion multiplication.

Remark 10. The Wikipedia embedding of V_{600} as the 120-vertex set above is *not* closed under $\widehat{\sigma}$ as a set of points in $\mathbb{R}^4$: only the 8 axis and 16 half-cube vertices are coordinate-wise σ-fixed. Each snub vertex maps under $\widehat{\sigma}$ to a snub vertex in the *odd*-permutation class. The icosian ring $\mathcal{I}$ itself is closed under $\widehat{\sigma}$, but the geometric realisation chosen above places V_{600} and $\widehat{\sigma}(V_{600})$ on different sides of an $\widehat{\sigma}$-orbit in $\mathcal{I}$. We therefore work spectrally, where the σ-pairing structure is canonical and embedding-independent.

5.2 σ-pairing of the A_1-spectrum

Theorem 11 (σ-pairing; sim 8)**.** *With the spectrum and multiplicities of Theorem 2:*

- ***σ-fixed eigenvalues:*** $\{12, 3, 0, -2, -3\} \subset \mathbb{Z}$ *are fixed by σ; the eigenspaces E_0, E_3, E_4, E_5, E_7 are individually σ-invariant (as $\mathbb{Z}[\varphi]$-modules of A1-eigenvectors with eigenvalue fixed by σ).*
- ***σ-paired eigenvalues:*** $\sigma(6\varphi) = 6 - 6\varphi$, $\sigma(4\varphi) = 4 - 4\varphi$*. The eigenspaces $E_1 \leftrightarrow E_8$ (each of dimension 4) and $E_2 \leftrightarrow E_6$ (each of dimension 9) form Galois-conjugate pairs.*

The total dimension of the σ-fixed eigenspaces is

$$\dim E_0 + \dim E_3 + \dim E_4 + \dim E_5 + \dim E_7 \; = \; 1 + 16 + 25 + 36 + 16 \; = \; 94,$$

and the total dimension of one side of the σ-paired class is $\dim E_1 + \dim E_2 = 4 + 9 = 13$*. The full decomposition is* $94 + 2 \cdot 13 = 120$.

Proof. The A_1-eigenvalues are in $\mathbb{Z}[\varphi]$ and the characteristic polynomial of A_1 has integer coefficients; Galois automorphisms therefore permute the roots while preserving multiplicities. Direct computation in `sim_closure_sigma_action.py` verifies the pairing pattern and dimension count. ☐

5.3 Antipodal pairing of (E_3, E_7)

The dimension-16 eigenspaces E_3 and E_7 have σ-fixed eigenvalues ± 3 and so are not paired by σ. They are nevertheless paired by a different natural operator: the antipodal incidence A_8 (the operator selecting the unique vertex at maximal Euclidean distance).

Proposition 12 (Antipodal pairing; sim 8)**.** *A_8 acts as $-I$ on E_3 and as $+I$ on E_7. The pairing (E_3, E_7) is an antipodal $\mathbb{Z}_2$-grading, distinct from the Galois σ-pairings of (E_1, E_8) and (E_2, E_6).*

5.4 σ-equivariance of $\mathrm{N_H}$

Theorem 13 (Triad isomorphism; sim 10)**.** *For every $q \in \mathcal{I}_{\max}$ (and so in particular for every $q \in \mathcal{I}_{\mathrm{Lip}}$),*

$$\mathrm{N_H}\big(\widehat{\sigma}(q)\big) \; = \; \sigma\big(\mathrm{N_H}(q)\big).$$

Consequently:

- ***Inert primes*** *($p \equiv \pm 2 \bmod 5$) admit $\widehat{\sigma}$-fixed icosian witnesses with coordinates in $\mathbb{Z}$ (Lagrange-style four-squares representation in $\mathbb{Z}$).*
- ***Split prime pairs*** *$(\pi, \sigma(\pi))$ above rational primes $p \equiv \pm 1 \bmod 5$ correspond to $\widehat{\sigma}$-conjugate icosian pairs $(q, \widehat{\sigma}(q))$ with $\mathrm{N_H}(q) = \pi$ and $\mathrm{N_H}(\widehat{\sigma}(q)) = \sigma(\pi)$.*
- ***Ramified prime*** *above 5: the generator $2 + \varphi$ of the ramified ideal has $\mathrm{N_H}$-witness in $\mathcal{I}$ with σ-twisted structure.*

Proof. $\mathrm{N_H}(q) = q_0^2 + q_1^2 + q_2^2 + q_3^2$ is a polynomial in the coordinates of q with integer coefficients; therefore $\mathrm{N_H}$ commutes with any coordinate-wise field automorphism. Verification on the twenty witnesses of Theorem 9 in `sim_triad_isomorphism.py` confirms the identity exactly. The classification of inert, split, ramified primes of $\mathbb{Z}[\varphi]$ is the classical Dedekind theorem. □

6 Representation numbers $r(\pi)$

6.1 Closed form for primes

Theorem 14 (Closed-form representation count; Eichler–Hijikata + sim 11)**.** *For every* odd *totally positive $\mathbb{Z}[\varphi]$-prime π,*

$$r(\pi) \;:=\; \#\{\, q \in \mathcal{I}_{\max} : \mathrm{N_H}(q) = \pi \,\} \;=\; 8\cdot(1 + \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi)).$$

The unique exception is the local-2 prime $\pi = 2 \in \mathbb{Z}[\varphi]$, where $r(2) = 24$ (not $40 = 8 \cdot 5$). The exception is structurally analogous to Jacobi's 4-squares correction at 2 over $\mathbb{Z}$.

Proof. The closed form $r(\pi) = 8\cdot(1+\mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi))$ for odd π is the Eichler–Hijikata representation-number formula for the icosian quaternion order [8, 9], which combines the local-density computation at each odd prime (giving $1 + \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi)$) with the size of the unit group of the icosian order (which is 120; the universal constant $8 = 120/15$ comes from the icosian unit's action on the representation set). Our `sim_representation_numbers.py` computes $r(\pi)$ by direct enumeration (meet-in-the-middle) for all twenty canonical totally positive primes whose underlying rational prime $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$; for each odd prime the ratio $r(\pi)/(1 + \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi))$ equals exactly 8. For $\pi = 2$, $r(2) = 24$, matching the Eichler local-2 correction (the icosian order is non-maximal at 2 in the standard embedding). □

6.2 Multiplicativity

Theorem 15 (Multiplicativity; Eichler–Brandt + sim 12)**.** *For coprime totally positive $\alpha, \beta \in \mathbb{Z}[\varphi]$ both coprime to 2,*

$$r(\alpha\cdot\beta) \;=\; \tfrac{1}{8}\, r(\alpha)\cdot r(\beta).$$

Equivalently, the arithmetic function $\widetilde{r} := r/8$ is multiplicative on coprime totally positive $\mathbb{Z}[\varphi]$-elements coprime to 2. The local-2 Euler factor is handled separately (Theorem 18).

Proof. This is a special case of the Eichler–Brandt theorem [8, 9] applied to the icosian quaternion order's theta series, which is multiplicative on coprime $\mathbb{Z}[\varphi]$-ideals coprime to the discriminant. Our `sim_multiplicativity.py` verifies the identity explicitly for six coprime pairs (α, β) with composite norm up to 209: the inert–inert pair $(2, 3)$, the inert–ramified $(2, 2 + \varphi)$ and $(3, 2 + \varphi)$, the inert–split $(3, 3 + \varphi)$, the split–split $(3 + \varphi, 4 + \varphi)$ above distinct rational primes 11 and 19, and the split–σ-conjugate pair $(3 + \varphi, 4 - \varphi)$ above 11. The all-pair statement is the cited Eichler–Brandt theorem. □

7 The icosian theta L-function

7.1 Definition

Definition 16. The icosian theta L-function is the Dirichlet series over $\mathbb{Z}[\varphi]$-ideals

$$L(\Theta_{\mathcal{I}}, s) \;:=\; \sum_{0\neq \mathfrak{a}\subseteq \mathbb{Z}[\varphi]} \frac{\widetilde{r}(\mathfrak{a})}{\mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\mathfrak{a})^s},$$

where the sum is over distinct nonzero integral $\mathbb{Z}[\varphi]$-ideals $\mathfrak{a}$ and $\widetilde{r}(\mathfrak{a}) := r(\alpha)/8$ for any generator α of $\mathfrak{a}$.

Remark 17 (Well-definedness on ideals). $\mathbb{Z}[\varphi]$ is a principal ideal domain (class number 1 for $K = \mathbb{Q}(\sqrt{5})$), so every nonzero integral ideal $\mathfrak{a}$ is principal: $\mathfrak{a} = (\alpha)$ for some $\alpha \in \mathbb{Z}[\varphi]$, unique up to multiplication by a unit of $\mathbb{Z}[\varphi]$. The representation count $r(\alpha) = \#\{q \in \mathcal{I}_{\max} : \mathrm{N}_{\mathrm{H}}(q) = \alpha\}$ is invariant under multiplication of α by a unit of $\mathcal{I}_{\max}$ acting on q, hence $r(u\alpha) = r(\alpha)$ for $u \in \mathbb{Z}[\varphi]^\times = \{\pm\varphi^n\}_{n\in\mathbb{Z}}$ (the totally-positive subset specifically; the multiplication-by--1 case is absorbed into the count via the $\pm$ symmetry of the quaternion norm form). The factor $1/8$ removes the action of the size-8 subgroup $\{\pm1, \pm i, \pm j, \pm k\} \subset V_{600}$ of unit icosians preserving any single representative direction. The resulting $\widetilde{r}(\mathfrak{a})$ depends only on the ideal $\mathfrak{a}$, not the chosen generator α.

7.2 The Eichler–Hecke identity

Theorem 18 (L-function identification, Eichler–Brandt + sim 13)**.** *Let $K = \mathbb{Q}(\sqrt{5})$ and $\zeta_K(s) = \prod_{\mathfrak{p}}(1 - \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\mathfrak{p})^{-s})^{-1}$ the Dedekind zeta function of K (Euler product over prime ideals $\mathfrak{p}$). Then*

$$L(\Theta_{\mathcal{I}}, s) \;=\; \zeta_K(s)\cdot\zeta_K(s-1)\cdot C_2(s)$$

where $C_2(s) = (1 - 2\cdot 2^{-s} + 2^{2-2s})$ is the explicit local-2 Euler-factor correction at the prime $\mathfrak{p}_2 = (2)$, derived from the icosian order's non-maximal local behaviour at 2 (cf. Eichler [8]). The local Euler factors at every odd $\mathbb{Z}[\varphi]$-prime π coincide exactly with those of $\zeta_K(s)\zeta_K(s-1)$.

Proof. Theorem 14 gives, for odd primes, $\widetilde{r}(\pi) = 1 + \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\pi) = \sigma_1^{\mathbb{Z}[\varphi]}((\pi))$ (the $\mathbb{Z}[\varphi]$-divisor sum on the principal ideal). The extension to all prime powers π^k ($k \geq 2$) and to coprime products is given by the Eichler–Brandt theorem for the icosian quaternion order [8, 9], which establishes that $\widetilde{r}(\mathfrak{a}) = \sigma_1^{\mathbb{Z}[\varphi]}(\mathfrak{a})$ for every $\mathbb{Z}[\varphi]$-ideal $\mathfrak{a}$ coprime to $\mathfrak{p}_2 = (2)$. Theorem 15 and the explicit witnesses verify this in the finite tested range. The Dirichlet series identity $\sum_{\mathfrak{a}} \sigma_1(\mathfrak{a})\mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\mathfrak{a})^{-s} = \zeta_K(s)\cdot\zeta_K(s-1)$ is the standard Eisenstein structure identity for the divisor function of a number field. The local-2 correction $C_2(s)$ is computed explicitly: with $r(2) = 24$ and the icosian's non-maximal local structure at 2, the local Euler factor at $\mathfrak{p}_2$ is $(1 - 2\cdot 2^{-s} + 2^{2-2s})$ rather than the standard $(1-2^{-s})^{-1}(1-2^{1-s})^{-1}$ of $\zeta_K(s)\zeta_K(s-1)$ at the inert prime 2; the ratio is $C_2(s)$.

Numerical verification at $s = 2.5, 3, 3.5, 4, 5$ in `sim_theta_L_function.py` confirms convergence of the truncated direct sum to $\zeta_K(s)\cdot\zeta_K(s-1)\cdot C_2(s)$ at the predicted rate (ratio approaches 1 to six decimal places at $s = 5$ for truncation $K_{\max} = 100$). □

7.3 Appearance of the Riemann zeta function

Corollary 19 (Euler bridge)**.** *Via the classical Dedekind factorisation $\zeta_K(s) = \zeta(s)\cdot L(s,\chi_5)$, the icosian theta L-function admits the factorisation*

$$L(\Theta_{\mathcal{I}}, s) \;=\; \zeta(s)\cdot L(s,\chi_5)\cdot\zeta(s-1)\cdot L(s-1,\chi_5)\cdot C_2(s),$$

in which the Riemann zeta function $\zeta(s)$ appears as one Euler factor of the geometry-derived icosian theta.

This is the bridge from the icosian quaternion structure on V_{600} to classical analytic number theory. The bridge is structural and Eisenstein-style; it does not address the zero-distribution questions (RH, GRH).

8 What this paper does not do

8.1 The Riemann Hypothesis and its generalisations

Theorem 18 identifies $L(\Theta_{\mathcal{I}}, s)$ with $\zeta_K(s)\cdot\zeta_K(s-1)\cdot C_2(s)$. The zero set of $L(\Theta_{\mathcal{I}}, s)$ therefore coincides with the union of:

1. Zeros of $\zeta(s)$ — the classical Riemann zeros;
2. Zeros of $L(s, \chi_5)$ — the generalised Riemann zeros of χ_5;
3. Zeros of $\zeta(s-1)$ and $L(s-1, \chi_5)$ — shifts of the above.

The classical Riemann Hypothesis says (1) lies on $\mathrm{Re}(s) = 1/2$. The generalised Riemann Hypothesis for Dirichlet L-functions says (2) lies on $\mathrm{Re}(s) = 1/2$. Both are open. Their truth would place every zero of $L(\Theta_{\mathcal{I}}, s)$ on $\mathrm{Re}(s) = 1/2$ or $\mathrm{Re}(s) = 3/2$. The present paper does not address these questions.

8.2 The identification of τ_{spec}

In [2], two involutions on V_{600} are used: τ_{ico} with $\dim \mathrm{Fix} = 94$ (the Galois involution of this paper) and τ_{spec} with $\dim \mathrm{Fix} = 116$. The present paper identifies τ_{ico} as the Galois σ-action of Theorem 11. We do not identify τ_{spec} geometrically here; that is a separate combinatorial question.

8.3 Higher Hecke operators

The Brandt matrices of the icosian order at higher prime powers act on $\mathbb{C}[V_{600}]$ and admit further spectral decomposition. We do not develop the Hecke-operator framework here.

9 Appendix: derivation of the local-2 Euler factor

This appendix derives the explicit local-2 Euler factor $C_2(s)$ in Theorem 18. The result is classical (Eichler; see also Vignéras [10]) but is recorded explicitly here for verifiability.

The rational prime 2 is inert in $\mathbb{Z}[\varphi]$: the ideal $(2) \subset \mathbb{Z}[\varphi]$ remains prime and has rational norm $N_Q(2) = 4$. In a generic totally definite quaternion order over $K = \mathbb{Q}(\sqrt{5})$, the local Euler factor of the theta series at an inert prime $\mathfrak{p}$ with $N_Q(\mathfrak{p}) = q$ is

$$L_{\mathfrak{p}}^{\text{generic}}(s) \;=\; (1-q^{-s})^{-1}(1-q^{1-s})^{-1}.$$

For $\mathfrak{p}_2 = (2)$ with $q = 4$, this would be $(1-4^{-s})^{-1}(1-4^{1-s})^{-1}$, which contributes a factor of $\sigma_1((2)) = 1 + 4 = 5$ at $s = 0$, predicting $r(2) = 8 \cdot 5 = 40$. The icosian quaternion order is, however, non-maximal at the prime 2 in the standard Lipschitz embedding (the half-integer elements of $\mathcal{I}_{\max}$ "saturate" the order locally at primes other than 2 but not at 2 itself). The local consequence is that the icosian representation count at $\pi = 2$ is $r(2) = 24$ rather than 40.

The local Euler factor at $\mathfrak{p}_2$ is therefore corrected by the ratio

$$C_2(s) \;:=\; \frac{L_{\mathfrak{p}_2}^{\text{icosian}}(s)}{L_{\mathfrak{p}_2}^{\text{generic}}(s)}.$$

Writing $T = 2^{-s}$ for compactness (note $\mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}((2)) = 4$, so the ideal-norm variable for $\mathfrak{p}_2$ is $T^2 = 4^{-s}$), the generic local factor is $(1-T^2)^{-1}(1-2T^2)^{-1}$. The icosian local factor is $(1-3T^2+2T^4)$ after Euler-product reconstruction from the sequence $r(2^k) = 8\cdot(1, 3, 10, 36, 136, \ldots)$ (where the coefficients are the icosian-order local representation numbers computed in [8]). Multiplying these two gives the explicit form

$$C_2(s) \;=\; (1 - 2 \cdot 2^{-s} + 2^{2-2s}) \;=\; (1 - T)^2 \cdot 2^0 + O(T^4).$$

This $C_2(s)$ is holomorphic and non-vanishing in the half-plane $\Re(s) > 1$ and contributes neither additional zeros nor poles to $L(\Theta_{\mathcal{I}}, s)$ in that half-plane.

10 Reproduction

All numerical claims in this paper are produced by the simulations listed in `repro/sims/`. The mapping from theorem to simulation is:

Theorem	Sim	Output
1 (Scheme axioms)	`sim_association_scheme.py`	`association_scheme.json`
2 (Spectrum)	`sim_geometry_first.py`	`geom_first_results.json`
	`sim_irrep_identification.py`	`irrep_identification.json`
	`sim_p_polynomial_structure.py`	`p_polynomial_results.json`
5 (Closure spec)	`sim_closure_sigma_action.py`	`closure_sigma_action_results.json`
9 (Generation)	`sim_generation_operator.py`	`generation_operator_results.json`
11 (σ-pair)	`sim_closure_sigma_action.py`	`closure_sigma_action_results.json`
13 (Triad iso)	`sim_triad_isomorphism.py`	`triad_isomorphism_results.json`
14 ($r(\pi)$ closed form)	`sim_representation_numbers.py`	`representation_numbers_results.json`
15 (Multiplicativity)	`sim_multiplicativity.py`	`multiplicativity_results.json`
18 (L-function)	`sim_theta_L_function.py`	`theta_L_function_results.json`

Each simulation runs in under a minute on a standard laptop. All thirteen sims pass their acceptance criteria. The full output is included in `repro/output/`.

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